

MicroFinance Core Banking & Lending Management System

A Comprehensive, End-to-End Enterprise Solution for Multi-Branch Microfinance Institutions, NBFCs, Mutual Benefit Trusts & Credit Cooperatives.



Multi-Branch Core Banking

Real-time Savings & Current accounts, inter-branch fund settlements, automated interest calculation, and digital passbooks.



360° Lending Lifecycle

Complete workflows for Personal Loans, Business Loans, Secured Gold Loans, and Joint Liability Groups (JLG).



Term Deposit & Policy Portfolio

Automated FD, RD, Daily Doorstep Deposits (DDS), MIS, and Flexible Recurring Investment management.



Dual-Control Maker-Checker

Fraud-proof multi-tier approval pipeline for KYC, loans, investments, vouchers, and cash transfers.

SOFTWARE SPECIFICATION & DELIVERABLES
DOCUMENT

Complete Operational & Technical Capabilities Guide

DOCUMENT STATUS

Client Presentation Version

2026

The **MicroFinance ERP System** is an industrial-strength financial operating platform engineered to automate every aspect of banking, micro-credit lending, member equity, and field agency operations. Built on proven Spring Boot enterprise technologies, it guarantees zero calculation discrepancies, complete double-entry ledger integrity, strict maker-checker governance, and real-time regulatory compliance reporting.

15+

CORE MODULES

100%MAKER-CHECKER
AUDIT**Multi**

BRANCH HIERARCHY

Real-Time

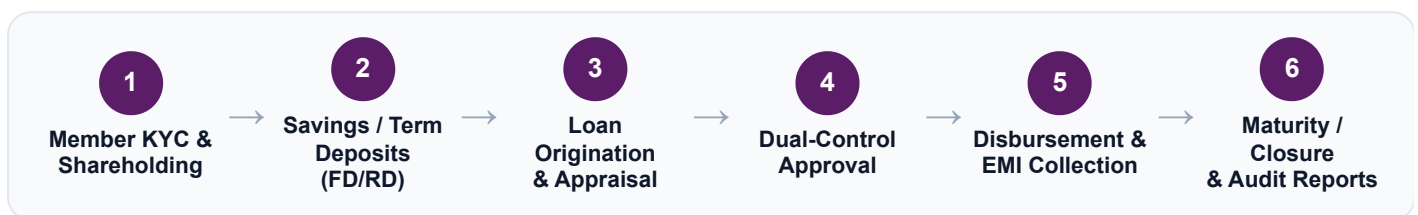
BALANCE SHEET & P&L

Technology & Security Architecture

ROBUST & SCALABLE

- ✓ **Backend Engine:** Java Spring Boot 2.7.x with enterprise JPA / Hibernate ORM.
- ✓ **Database & Data Integrity:** Relational ACID compliance with foreign key cascades and automated transaction rollbacks.
- ✓ **Security & RBAC:** Granular Role-Based Access Control, session timeouts, and isolated module-menu access permissions.
- ✓ **Report Automation:** Apache POI dynamic Excel exports and print-ready thermal/A4 vouchers and legal bonds.

End-to-End Operational Lifecycle Workflow



⚙️ Maker-Checker Dual Governance Framework

No high-value transaction, member onboarding, loan sanction, or maturity disbursement takes effect immediately upon entry. Field operators create requests that are routed to branch managers/auditors in real-time approval queues, preventing internal leakage.

1. Preferences & Multi-Branch System Setup

SYSTEM FOUNDATION

Centralized configuration suite governing institution identity, organizational hierarchy, banking partners, and security policies across all branches.

- **Company Administration:** Set institution legal name, registration numbers, CIN/TAN, logo, and letterhead credentials.
- **Multi-Branch Hierarchy:** Create unlimited physical branches, sub-offices, and regional hubs with unique prefix codes.
- **Dynamic Numbering (Codes):** Auto-generate customized serial numbers for Members, Accounts, Policies, and Loans.
- **Financial Year Control:** Open, lock, and transition fiscal periods with ledger balances carried forward safely.
- **Bank & Account Masters:** Configure commercial bank accounts, clearing account numbers, and IFSC/branch tags.
- **Granular User & Menu RBAC:** Assign fine-grained view/edit/delete access rights per menu per user role.

2. Customer 360 & KYC Management

MEMBER LIFECYCLE

Complete digitization of customer relationship onboarding, document verification, biometric/photo archives, and unified financial history.

- **Digital Member Onboarding:** Capture full demographic details, guardian links, occupation, category, and nominee records.
- **Customer 360° Summary:** Single-window dashboard displaying all active savings, FD/RD policies, and loans under one member ID.
- **Integrated KYC Vault:** Upload and verify Aadhaar Card, PAN Card, Voter ID, Ration Card, and Member Signature/Photo.
- **Advanced Member Search:** Instant lookup by Customer No, Mobile Number, Aadhaar Number, Name, or Branch.

3. Customer Shareholding & Equity Management

STATUTORY COMPLIANCE

Mandatory statutory module for Mutual Benefit & Credit Societies to manage member equity capital, distinctive numbers, and certificates.

- **Share Allotment Engine:** Issue equity shares upon membership enrollment with face value and share quantity tracking.
- **Share Certificate Generation:** Print professional, tamper-proof legal Share Certificates with seal and signature placeholders.
- **Share Transfer System:** Regulated transfer of shares between existing members with board resolution approval flags.
- **DNO Tracking & Unallotted Shares:** Distinctive Number series tracking with live unallotted equity inventory reports.

4. Savings & Current Account (Core Banking)

CORE BANKING ENGINE

Full-featured deposit operations supporting cash counters, passbook printing, fund transfers, and automated interest posting.

- **Account Opening & Scheme Catalog:** Configure minimum balance, interest rates, nomination, and joint operating rules.
- **Internal Fund Transfers:** Instant real-time fund transfers between internal savings accounts with auto-contra entries.
- **Savings Passbook & Statements:** Print chronological passbook updates and export statement sheets to PDF/Excel.
- **Deposit & Withdrawal Counter:** Multi-channel deposit and withdrawal processing with cash verification and receipt printing.
- **Automated Interest Engine:** Calculate and post interest on daily product or quarterly average balances with TDS rules.
- **Account Closure & Maintenance:** Formal account closing workflow, unfreeze options, and annual SMS service charge deductions.

5. Policy & Term Deposit Management (FD, RD, MIS, DDS)

INVESTMENT ENGINE

Industrial deposit plan catalog enabling micro-savings schemes, daily doorstep collections, and high-yield fixed investments.

Deposit Scheme	Operational Mechanism	Key Features & Controls
Fixed Deposit (FD)	One-time lumpsum deposit with compounding tenure interest.	Auto-calculated maturity value, bond generation, premature withdrawal penalties.
Recurring Deposit (RD)	Monthly fixed installment savings over predetermined months.	Late payment penalty calculation, installment passbooks, payment receipts.
Daily Deposit (DDS / Pigmy)	Doorstep daily small savings collected via field agents.	Batch daily premium entries, agent collection tracking, flexible tenures.
Monthly Income Scheme (MIS)	Lumpsum deposit with monthly interest payouts to savings account.	Automated monthly interest disbursement register, reinvestment options.
Flexible Premium Plan	Variable investment amounts with flexible renewal schedules.	Customized interest compounding, transaction slips, and partial liquidation.

- **Bond / Certificate Generation:** Print official Investment Bonds and Bond Re-issuance logs.
- **Installment Record Book:** Micro-level ledger tracking every renewal installment and penalty waiver.

6. Standard & Retail Loan Management

PERSONAL & BUSINESS CREDIT

End-to-end loan origination, risk appraisal, amortization scheduling, disbursement, collection, and recovery management.

- **Scheme Catalog & EMI Calculator:** Configurable Flat and Reducing balance interest schemes with interactive repayment simulators.
- **Multi-Stage Loan Sanction:** Committee review, document checklist verification, and formal disbursement via cash or bank.
- **Sanction Letters & Loan Ledger:** Auto-generate formal loan agreements, repayment passbooks, and real-time principal balance ledgers.
- **Application & Co-Signer Appraisal:** Capture loan purpose, guarantor details, asset collateral, and credit sanction terms.
- **Regular & Irregular EMI Collection:** Accept scheduled EMIs, advance installments, or irregular partial payments with fine tracking.
- **Foreclosure, Settlement & NOC:** Early closure interest rebate calculation, loan settlement, and automated No Objection Certificates.

7. Secured Gold Loan Management

COLLATERAL BACKED

Specialized asset-backed lending module with precision karat valuation, locker safekeeping, and pledge custody tracking.

- **Gold Directory & Valuation:** Record ornament types, gross weight, stone deduction, net weight, purity karats, and market rate LTV.
- **Gold Loan Sanction & Agreements:** Generate gold pawn tickets, customer pledge agreements, and disbursement vouchers.
- **Safe Vault & Locker Assignment:** Assign secure locker packet numbers, appraiser certification, and tamper-proof bag seals.
- **Pledge Release & Gold NOC:** Complete redemption protocol upon full settlement with collateral handover sign-off and Gold NOC.

8. Joint Liability Group (JLG) & Micro-Credit

GROUP LENDING MODEL

Micro-credit group lending engine based on Grameen / SHG models with mutual peer guarantees and center meeting collections.

- **Lending Group & Center Formation:** Create Self-Help / JLG groups (5-20 members), assign Group Leader, and link to collection centers.
- **Center Meeting Repayment Portal:** Bulk collection sheet processing for weekly/bi-weekly center meetings with one-click posting.
- **Group Loan Application:** Mutual guarantee agreement, cross-liability tracking, and unified group loan approval.
- **Group Performance & Settlement:** Group statement of accounts, default resolution, and completed group loan archival.

9. Dual-Control Maker-Checker Approval System

ZERO FRAUD SECURITY

Industry-standard four-eye principle ensuring all financial operations and master modifications undergo independent verification.

Approval Queue	Workflow Scope	Risk Prevention Goal
Customer & KYC Approvals	Verifies identity proofs and member eligibility.	Prevents ghost accounts and duplicate registrations.
Share & Equity Transactions	Validates share transfers and certificate allocations.	Ensures strict equity ledger balancing.
Investment Schemes (FD/RD/DD)	Authorizes new term deposit booking & rate locks.	Avoids unauthorized interest rate offerings.
Savings & Account Transfers	Validates high-value deposits, cashouts & fund transfers.	Eliminates teller counter discrepancies.
Loan Sanctions & EMI Payments	Supervises credit approvals and manual waiver postings.	Stops unauthorized disbursement & fine waivers.
Maturity & Inter-Branch Cash	Clears final claim payouts and physical cash movements.	Guarantees cash vault and fund security.

10. Regulated Data Correction & Rectification Engine

SUPER-ADMIN AUDIT

Supervised administrative correction portal with tamper-proof logging to rectify legitimate operator errors without breaking ledger chains.

- **Member & Policy Correction:** Correct name typos, addresses, and renewal date misentries with change-history logs.
- **Financial Entry Adjustment:** Reverse and rectify accidental journal debits/credits without corrupting historical balance sheets.
- **Transaction Removal with Audit:** Super-admin controlled deletion of erroneous flexible/savings records with reason logging.

11. Maturity Management & Payout Processing

CLAIM SETTLEMENTS

Comprehensive policy maturity engine managing claim lodgments, interest compounding calculations, and payout execution.

- **Maturity Scheme Rules:** Pre-program tenure, maturity bonus, loyalty interest additions, and lock-in periods.
- **Claim Lodgment & Tracking:** Submit maturity applications upon policy term completion with live status monitoring.
- **Full & Partial Maturity Settlements:** Automated deduction of premature penalties or payout of full principal + interest via Bank/Cheque.

12. Financial Consultant & Agency Hierarchy (MLM Commission Engine)

FIELD SALES
GROWTH

Multi-tier agency relationship and commission engine to motivate field advisors, manage downlines, and auto-calculate incentives.

- **Advisor Onboarding & Hierarchy:** Register financial advisors, assign senior introducers, and maintain unlimited downline trees.
- **Downline Tree Visualization:** Visual hierarchical tree mapping showing active advisors, rank cadences, and sales volume.
- **Dynamic Incentive Scheme Master:** Define tiered commission percentages for FD, RD, Daily deposits, and loan originations.
- **Commission Billing & ID Cards:** One-click generation of monthly advisor payout slips, summary reports, and printable Photo ID cards.

13. Human Resources & Team Management

WORKFORCE CONTROL

Internal staff governance suite for defining departmental hierarchies, designation privileges, and automated ID issuance.

- **Department & Designation Masters:** Structure administrative departments (Operations, Audit, Credit, Cash) and ranks.
- **Team Member Onboarding:** Maintain staff bio-data, branch assignment, joining dates, and emergency contacts.
- **Employee ID Card Generator:** Automated generation of professional employee identification cards with company credentials.

14. Complete Financial Accounting & Banking Operations

STATUTORY DOUBLE-ENTRY

Full-scale financial accounting suite delivering flawless double-entry ledgers, automated bank reconciliation, and real-time statements.

- **Chart of Accounts (Ledger Master):** Hierarchical grouping of Assets, Liabilities, Incomes, and Expenses with opening balances.
- **Cheque Clearance & Contra Transfers:** Cheque deposit registers, clearance/bouncing workflows, and Cash-Bank contra entries.
- **Cash Book & Day Book:** Multi-column real-time Cash Book and Daily Transaction Daybook with opening/closing vault reconciliation.
- **Vouchers (Receipt, Payment & Journal):** Outgoing expense vouchers, incoming revenue receipts, and double-entry manual journals.
- **Inter-Branch Cash Transfers:** Regulated physical cash transit between branch vaults with transit ledger validation.
- **Trial Balance, P&L & Balance Sheet:** Real-time automated generation of Trial Balance, Profit & Loss, and Balance Sheet.

15. Reports & Business Intelligence Suite

20+ REAL-TIME REPORTS

Comprehensive reporting engine offering instant operational insight, loan recovery analytics, and audit readiness.

-  **Investment Transaction Reports:** Date-wise, branch-wise, and scheme-wise investment collection summaries.
-  **Recurring Installment Due Report:** Proactive identification of unpaid RD and deposit installments for field agents.
-  **Upcoming EMI Due & Default Analysis:** Real-time monitoring of upcoming installments and overdue NPA default aging.
-  **Maturity Obligation Forecasting:** Project future liability obligations across upcoming months for liquidity planning.
-  **Advisor Performance Reports:** Individual and team collection summaries for monthly incentive disbursements.
-  **Loan Disbursement & Payment Reports:** Detailed breakdown of sanctioned capital, recovery amounts, and interest yields.
-  **Legal & Confirmation Documents:** Instant generation of Loan Sanction Letters, Gold Pawn Slips, JLG Agreements, and NOCs.
-  **Cheque Clearance Tracking:** Audit log of pending, cleared, and dishonored cheques across all bank accounts.

Client Package & System Deliverables Summary

Deliverable Item	Specification & Format	Status
Core ERP Software Source & Binary	Java Spring Boot Enterprise Application (WAR / Executable)	Fully Functional
Database Schema & Seed Data	Relational MySQL / MariaDB schema with automated tables	Included
Document & Bond Generators	Built-in PDF/HTML engines for Certificates, Passbooks & ID Cards	Integrated
Maker-Checker Multi-Tier Security	14+ Real-time approval queues with Role-Based Access Control	Active
Statutory Accounting Engine	Double-Entry General Ledger, Daybook, P&L, Trial Balance & Balance Sheet	Standard

MicroFinance Enterprise Banking & Lending Management System

Engineered for High Performance, Multi-Branch Security & Complete Regulatory Compliance.

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